

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 1

Case Study: Modelling Daily Exchange Rate Returns

Application area: *Finance — Foreign Exchange*

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Yahoo Finance (<https://finance.yahoo.com>): download daily USD/BRL (BRL=X) and EUR/USD (EURUSD=X) closing prices, 2000–2024, via `getSymbols()` in R (`quantmod`) or `yfinance` in Python.

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1. Background and Objectives

The daily USD/BRL exchange rate is one of the most volatile among emerging-market currencies, reflecting Brazil's commodity dependence, political cycles, and external financing conditions. This tutorial applies the Box–Jenkins ARMA framework to log-returns and evaluates whether simple time series models can capture the dynamics of this series.

Learning objectives: (i) Download and prepare financial time series data; (ii) apply stationarity tests and ACF/PACF identification; (iii) estimate and validate ARMA models; (iv) produce out-of-sample forecasts and evaluate accuracy.

2. Data Preparation

```
library(quantmod); library(tseries); library(forecast)

# Download USD/BRL from Yahoo Finance
getSymbols("BRL=X", src="yahoo", from="2000-01-01", to="2024-12-31")
brl <- na.omit(Cl('BRL=X'))
log_br1 <- log(brl)
ret_br1 <- diff(log_br1)[-1] # log-returns
```

```
import yfinance as yf, numpy as np, pandas as pd
brl = yf.download("BRL=X", start="2000-01-01", end="2024-12-31")["Close"]
log_br1 = np.log(brl.dropna())
ret_br1 = log_br1.diff().dropna()
```

3. Exercises

3.1. Descriptive analysis

Plot the log USD/BRL level and the log-returns. Compute and report: mean, standard deviation, skewness, and excess kurtosis of the returns. What do these statistics tell you about the distribution?

Excess kurtosis > 0 confirms fat tails. Negative skewness would indicate larger negative returns. These features motivate GARCH modelling in Tutorial 6.

3.2. Stationarity

- (a) Apply the ADF test (with constant, BIC lag selection) to the log level. Interpret the result.
- (b) Apply the ADF test to the log-returns. What do you conclude about the integration order?
- (c) Apply the KPSS test to both series. Do the conclusions agree?

3.3. ACF/PACF identification

Plot the ACF and PACF of the log-returns up to lag 20. Based on the plots, suggest two or three candidate ARMA(p, q) specifications. Explain your reasoning.

3.4. Estimation and model selection

- (a) Estimate your candidate models by MLE.
- (b) Compute AIC and BIC for each. Which model is selected by BIC?
- (c) Inspect the roots of the selected model. Are all characteristic roots outside the unit circle?

3.5. Validation

For the BIC-selected model:

- (a) Apply the Ljung–Box test to residuals (at lags 5, 10, 20). Is there evidence of remaining serial correlation?
- (b) Apply the Ljung–Box test to squared residuals. What does this tell you?
- (c) Plot CUSUM and CUSUMSQ of recursive residuals. Is there evidence of structural instability?

The Ljung–Box test on squared residuals should reject white noise — evidence of ARCH effects that motivate volatility modelling in Tutorial 6.

3.6. Forecasting

Re-estimate the model on 2000–2023. Produce 1-step-ahead forecasts for the full year 2024. Compute RMSFE, MAE, and Theil's U relative to a random walk benchmark. Does the ARMA model outperform the random walk?

4. Core References

Box, G.E.P. et al. (2015). *Time Series Analysis*, 5th ed. Wiley. Fama, E.F. (1965). *Journal of Business*, 38, 34–105. Lo, A.W. & MacKinlay, A.C. (1988). *Review of Financial Studies*, 1, 41–66.

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